

Group Report: Green Energy Data Analysis

Group Number: 36

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Contributions:

- 24280066: Developed Reddit data collection script, integrated Google Trends API
- 24280062: Processed public energy datasets, created pipelines

Github link: <https://github.com/24280066/Assignment1>

Overview of The Topic

Why Green Energy?

We selected "Green Energy" for this assignment. It was selected due to its global importance and its relevance for Pakistan. This topic is discussed on various forums but a data led approach may be very useful to prepare action items.

We had following three sources for data

- Reddit: Discussions about solar/wind energy challenges and adoption
 - Google Trends: Seasonal interest patterns in renewable technologies
 - Public Data: Kaggle, CSV was exported.
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Data Collection Process and Initial observations

As was guided, we tried Reddit Collection (praw). It was used for green energy discussion and we had put limit of 50. For reddit an app was created, through which a personal script and key was obtained. Then data was accessed through API. Collects posts from renewable energy subreddits The search words are in the code, they included renewable energy, solar, wind energy. We also removed deleted, empty content. Please refer to the output below;

```
Cleaning data...
Saving to CSV...
Reddit data saved to ./datasets/raw/reddit_posts.csv
Summary:

Total posts: 53
Subreddit distribution:
- RenewableEnergy: 16 posts
- solar: 31 posts
- windenergy: 6 posts
```

Next, we tried Google Trends (via pytrends) for search interest data. We had put time frame for it and days. It tracks keyword trends over 12 months Partial data was removed. The keywords were kept the same as solar power', 'wind energy', 'renewable energy'. Interest score was being gauged through it. Please refer to the output below;

```
Google Trends Summary:
- Date Range: 2024-02-11 00:00:00 to 2025-02-09 00:00:00
- Keywords: ['solar power', 'wind energy', 'renewable energy']
- Average Interest Scores:
  - renewable energy: 79.1
  - solar power: 78.4
  - wind energy: 29.5
```

For Public Dataset: we visited Kaggle for energy statistics however we were not able to connect through an API, so we downloaded world energy overview data in CSV and then handled it through the code.

It was structured energy data in columns, spread over years. Please refer to the output below;

```
Energy Trends (1973-2022)
=====
Peak Renewable Production: 1,218,790 units
Average Fossil Fuel Usage: 5,034,634 units/year

Updated data saved to Google Drive
```

AI Product Proposal

Green Energy Dashboard:

- Live sentiment analysis from Reddit
- Trend prediction using Google search data
- Energy production forecasts

Legal Considerations

Reddit:

- Storing personal information including usernames.

Google Trends:

- Cannot redistribute raw trend data
- Results limited to 5-year historical window

Kaggle, government data:

- Sensitive information and sharing limitations.

Multi-Source Analysis

The challenges are that the data formats differ and for time series the data may not be accurate for the reporting period. Metrics / unit of measurement are different.

Data Storage Plan

Structure:

Copy

/datasets/

├── raw/

| ├── reddit_posts.csv

| └── google_trends.csv

└── combined_energy.csv

└── processed/

└── combined_energy.csv

Combination Strategy:

python

Copy

```
merged_df = pd.merge(
```

```
    trends_data,
```

```
    energy_data,
```

```
    on='year'
```

```
)
```

Data Flow and Pipeline:

- API/Public Data → Fetch → Clean → Save
- Separation of raw vs processed data
- Scripts follow lab1 pattern (fetch/clean/save functions)

Output

Current Structure:

datasets/

processed/

cleaned_energy_data.csv

raw/

world_energy_raw.csv

reddit_posts.csv

google_trends.csv